Household Electricity Forecaster

- Forecasts a single household's electricity use **hour by hour for the next 24 hours** using Python, pandas and scikit-learn.
- Includes a time-based evaluation against simple baselines and an honest look at where the model does and does not help.

Data

Individual Household Electric Power Consumption (UCI Machine Learning Repository): this file records the electricity consumption of a household in Paris, one reading per minute from December 2006 to November 2010 (about 2 million rows and 1.25% are missing).

This file can be downloaded from the link above (not included in the repository).

Method

1- Averaged the minute readings into hourly power(kW): which gave me 34,589 hours with 421 of them missing (not filled in)
2- Built features known: hour of day, day of week, month and the value 24 hours and one week earlier
3- Time-based split: train up to 31 May 2010 and test from 1 June 2010
4- Compared 2 baselines and 2 models,

Results (hourly MAE, test period)

Method	MAE (kW)
Same hour yesterday (best baseline)	0.486
Same hour last week	0.541
Linear regression	0.473
Gradient boosting	**0.426**

Gradient boosting has about 12% lower hourly error than the best baseline

Average power in test period = 0.927kW

Limitations

- **Daily totals are not better than the baseline.** Over 156 complete test days, the model's average miss was 4.20 kWh/day versus 4.43 for "yesterday's total", and it was closer on only 50.6% of days. The model's value is the shape of the day, not the total.
- The model is worst in the evening, when usage is busiest and spikes are hard to time.
- One household is noisy, and there is no weather, holiday or occupancy information.
- Features use values from 24 hours and 7 days earlier, so the forecast horizon is up to 24 hours.

Run it with pip install –r requirements.txt

Open the notebook, point it at household_power_consumption.txt, and run all cells. Use forecast_day("2010-07-15") to plot a forecast for any day from 1 June 2010 onward (earlier days are in the training data).

Credit

Hebrail, G. and Berard, A. (2006). *Individual Household Electric Power Consumption* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C58K54 (CC BY 4.0)